

Fraud Transaction Analysis

1. Project Overview

This project focuses on building a Fraud Detection Dashboard using Power BI Desktop connected to a PostgreSQL database.

The goal is to:

Identify fraudulent transactions

Analysis fraud patterns

Support quick decision-making for fraud analysts.

2. Data Source & Setup

Database: PostgreSQL

Table: transactions

Created a SQL View: fraud dashboard

Why View?

Simplifies data for reporting

Improves performance

Industry best practice.

3. SQL Logic Used

```
CREATE VIEW fraud_dashboard AS
```

```
SELECT
```

```
    txn_id,
```

```
    user_id,
```

```
    amount,
```

```
    txn_type,
```

```
    location,
```

```
    is_fraud,
```

```
    EXTRACT(HOUR FROM txn_time) AS txn_hour
```

```
FROM transactions;
```

Explanation:

Extracted txn_hour → used for time-based fraud analysis

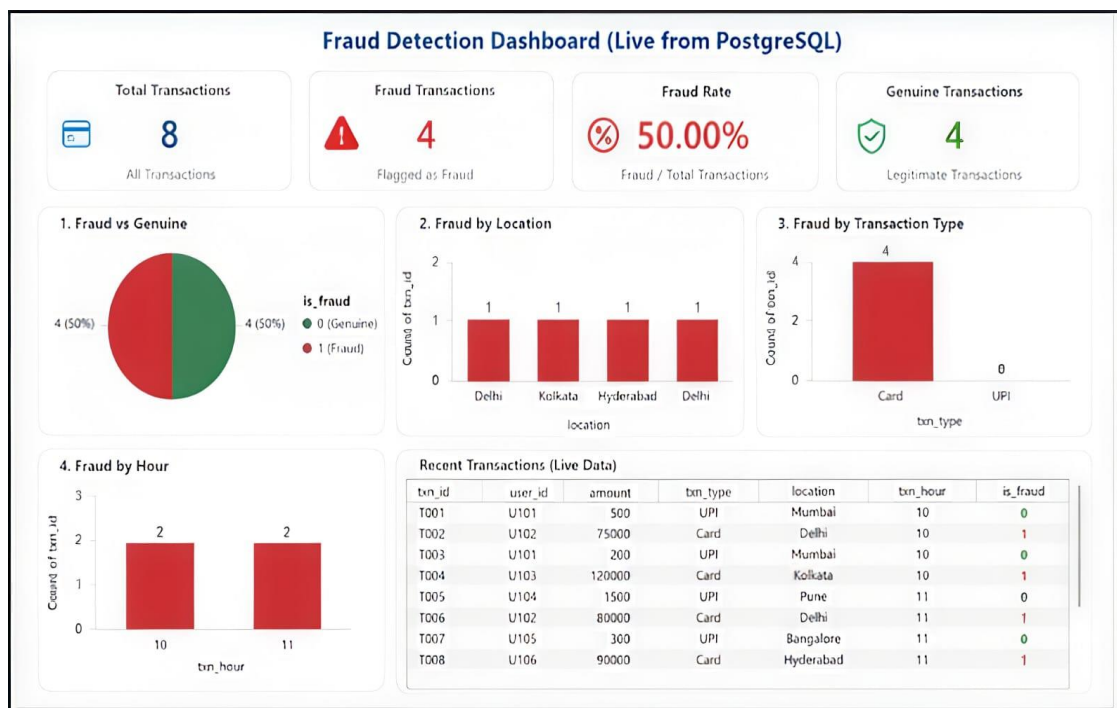
Selected only required fields → optimized dashboard

4. DAX Measure Used

Fraud Rate = $\text{DIVIDE}(\text{SUM}(\text{is_fraud}), \text{COUNT}(\text{txn_id}))$

Used to calculate fraud percentage for monitoring performance.

5. Dashboard Visuals Explanation



- Fraud vs Genuine

Shows proportion of fraud vs normal transactions

Helps understand overall fraud percentage

- Fraud by Location

Displays cities with high fraud

Useful for geo-risk analysis

- Fraud by Transaction Type

Compares Card vs UPI fraud

Helps identify risky payment channels

- Fraud by Hour

Shows time when fraud occurs most

Helps in real-time monitoring & alert rules

- Transaction Table

Detailed view of all transactions

Used for investigation

6. Key Metrics (KPIs)

- Total Transactions → Total number of transactions
- Fraud Transactions → Transactions flagged as fraud
- Fraud Rate → % of fraud transactions
- Genuine Transactions → Non-fraud transactions

These KPIs provide a quick risk snapshot.

7. Business Insights

- Card-based transactions show higher fraud exposure
- Certain locations exhibit repeated fraud patterns
- Fraud activity peaks at specific hours, enabling alert optimization.

8. Project Value

This dashboard helps:

- Reduce manual investigation effort
- Enable faster fraud detection
- Support real-time monitoring
- Improve decision-making for fraud analysts

9. Tools & Technologies

- Power BI
- PostgreSQL
- SQL
- DAX

10. Conclusion

This project demonstrates how data analytics can be used to:

- Detect fraud trends
- Improve monitoring systems
- Support proactive fraud prevention